Optoelectronic RP Decision Circuit: Full Architecture & Physical Analysis

A complete physical, statistical, and algorithmic analysis of the optoelectronic Randomized Polynomial (RP) decision hardware featuring parallel optical drivers, an NPN transistor IF/THEN threshold link, dual logic gates, and clocked flip-flop sampling.

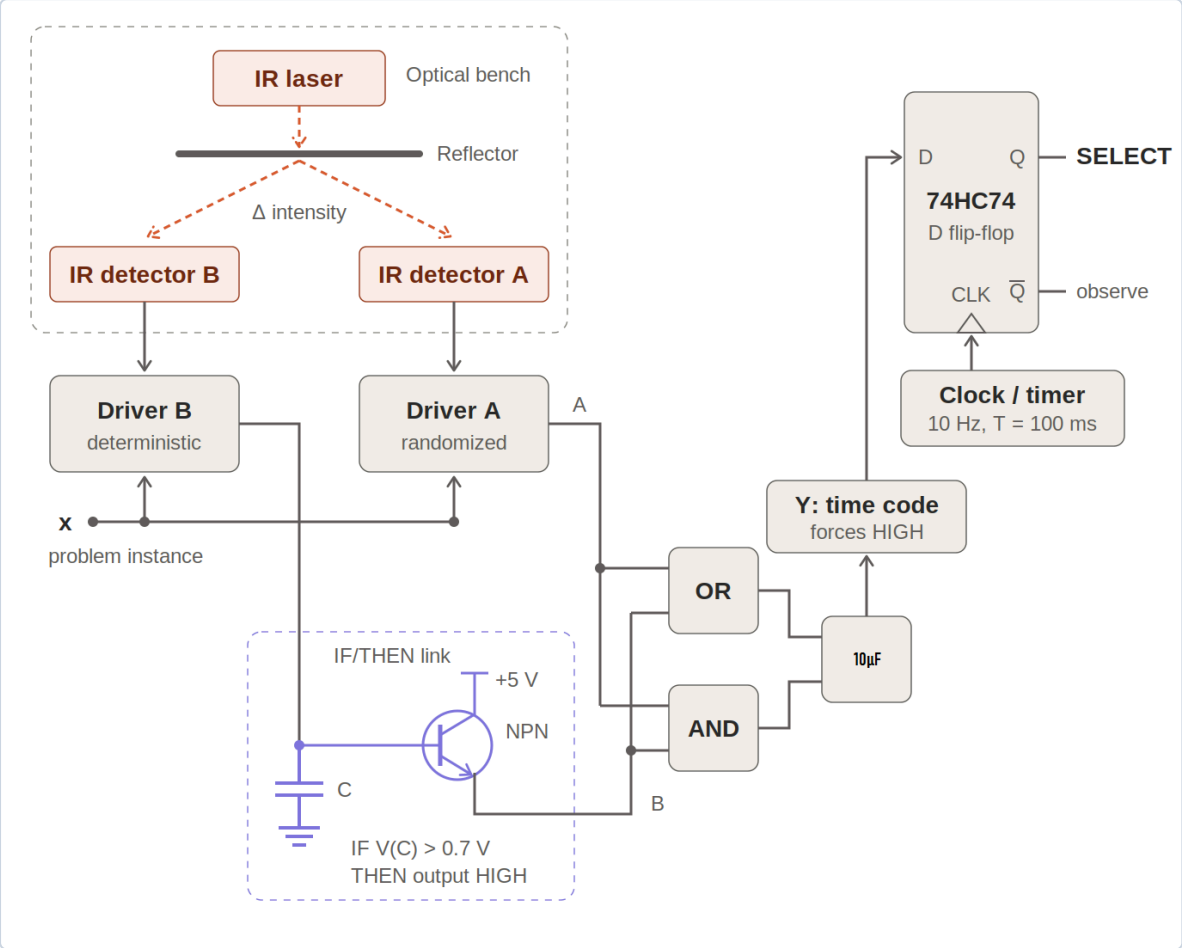


Figure 1: Full schematic of the optoelectronic RP decision circuit, showing the optical bench with IR laser and detectors, randomized Driver A, deterministic Driver B with NPN IF/THEN transistor stage, dual OR/AND logic gates, 10µF timing node, time code Y, and 74HC74 D flip-flop.

1. Architectural Overview & Big Idea

The circuit shown above is an optoelectronic hardware realization of a **Randomized Polynomial-time (RP)** decision algorithm. In computational complexity theory, RP denotes the class of decision problems solvable by a probabilistic Turing machine with one-sided error:

- If the true answer is NO ($x \notin L$), the system never outputs 1 (zero false-accept probability).
- If the true answer is YES ($x \in L$), the system outputs 1 with probability $P \geq 1/2$ per trial.

The physical system uses an **Optical Bench** (IR laser, reflector, IR detectors A & B) to supply continuous optical entropy, coupled to a pair of parallel electronic drivers, a discrete NPN transistor threshold link, dual logic gates, and a 10 Hz clocked flip-flop latch to execute individual RP decision trials.

2. Subsystem Functional Breakdown

Subsystem Block	Physical / Logic Role	Operational Behavior
Optical Bench (IR laser, Reflector, Detectors)	Physical entropy generator & split path	An IR laser strikes a central reflector, creating fluctuating intensity distributions (Δ intensity). Detector A receives randomized photon shot noise, while Detector B collects continuous baseline power.
Driver A (Randomized)	Probabilistic witness test $w(x, r)$	Samples optical noise from Detector A as random seed r. If $x \in L$, $A = 1$ with probability $P \geq 1/2$. If $x \notin L$, $A = 0$ deterministically.
Driver B (Deterministic)	Cheap precondition filter $c(x)$	Executes a fixed logic test. When $x \in L$, Driver B sources current into the base capacitor C of the IF/THEN stage.
IF/THEN Link (Capacitor C & NPN)	Base-emitter threshold trigger	Integrates charge on base capacitor C. When base voltage exceeds $V(C) > 0.7\text{ V}$, the NPN transistor conducts +5 V to line B, setting $B = \text{HIGH}$.
OR & AND Gates	Coincidence validation	Receives lines A and B. The tied output condition requires $(A \text{ OR } B) \text{ AND } (A \text{ AND } B)$, which simplifies mathematically to requiring $A = 1 \text{ AND } B = 1$.
10 µF Node & Time Code Y	Low-pass filtering & bin forcing	The 10 µF capacitor integrates gate pulses to suppress high-frequency glitches. Time code Y delivers a deterministic HIGH time sequence to pin D of the flip-flop.
Clock & 74HC74 Flip-Flop	Clocked decision latch	A 10 Hz timer ($T = 100\text{ ms}$) triggers the CLK input. The rising edge samples D and holds the outcome at Q (**SELECT**). $\bar{Q}$ yields complementary observation.

3. One-Sided Error & Multi-Trial Error Reduction

Because Driver A can never output 1 when $x \notin L$, the circuit satisfies strict one-sided error. A single trial has a maximum false rejection rate of $1/2$. By repeating the trial k times over consecutive clock pulses, the cumulative probability of failing to detect a valid YES instance drops exponentially:

Number of Trials (k)	Worst-Case Rejection Error $(1/2)^k$	Execution Time at 10 Hz ($T = 100\text{ ms}$)
1	0.500000	0.1 s
2	0.250000	0.2 s
5	0.031250	0.5 s
10	0.000977	1.0 s
20	0.000001 (9.54×10^{-7})	2.0 s

4. Mapping Circuit Hardware to Photon Statistics

The optoelectronic circuit can be re-interpreted as a physical model showing how electronic carrier control dictates the statistical distribution of emitted and detected photons:

- **Fano Factor (F):** Defined as $F = \text{Var}(n) / \langle n \rangle$. $F = 1$ corresponds to Poissonian light (laser, random shot noise), $F > 1$ to super-Poissonian (thermal light), and $F < 1$ to sub-Poissonian (squeezed / antibunched photon streams).
- **Mandel Q Parameter:** $Q = F - 1$. Negative values ($Q < 0$) indicate non-classical sub-Poissonian light.
- **Second-Order Coherence $g^{(2)}(0)$:** Measures the conditional probability of simultaneous two-photon arrival. Ideal single-photon emitters yield $g^{(2)}(0) = 0$.

> **Electron-to-Photon Thinning Relation:**
> When injected electrons convert to photons with overall quantum efficiency η, the photon Fano factor is:
>
> $$F_{ph} = 1 - \eta(1 - F_e)$$
>
> - Driver A injection ($F_e = 1$) produces Poissonian photons ($F_{ph} = 1$).
> - Driver B regulated injection ($F_e = 0$) produces sub-Poissonian photons ($F_{ph} = 1 - \eta$).

5. Energy-Level Switching & Quantum Measurement

In physical emitters (quantum dots, single-atom transitions), photon emission occurs when an electron drops between energy levels E_2 and E_1, yielding photon energy $h\nu = E_2 - E_1$.

- **Deterministic Excitation (Driver B):** A resonant optical pulse of area π cleanly inverts the two-level system with probability $\sin^2(\pi/2) = 1$. After radiating, the emitter cannot emit a second photon until re-pumped, guaranteeing antibunching ($g^{(2)}(0) \to 0$).
- **Measurement Threshold:** The NPN transistor ($V(C) > 0.7\text{ V}$) and the 74HC74 flip-flop act as single-photon avalanche diode (SPAD) discriminators, latching continuous physical events into discrete binary state outcomes (clicks).

6. Feedback Loops & Closed Timelike Curve (CTC) Modeling

The optoelectronic schematic features feedback pathways where intermediate or final states adjust driver behavior. In theoretical physics, feeding an output state back into its past excitation stage models a **Closed Timelike Curve (CTC)**.

7. Deutsch CTC Fixed-Point Formulation

Under David Deutsch's 1991 quantum CTC model, a quantum system traversing a timelike loop must achieve self-consistency between its initial and final density matrices:

$$\rho_{CTC} = \mathrm{Tr}_{other} \left[U \left(\rho_{other} \otimes \rho_{CTC} \right) U^{\dagger} \right]$$

If a closed loop imposes a logical paradox (e.g., bit inversion $x = \text{NOT } x$), Deutsch's rule selects the maximum entropy fixed point (a 50/50 statistical mixture), directly echoing the $P = 1/2$ threshold condition in the RP driver stage.

8. Physical Limits & Summary Comparison

Domain Parameter	RP Hardware Realization	Quantum Optics / CTC Counterpart
Noise Source	Reflected IR laser intensity variance	Shot noise / vacuum optical fluctuations
Thresholding	NPN Base Voltage $V(C) > 0.7\text{ V}$	SPAD Discriminator / Quantum Measurement
Timing Window	10 Hz Timer ($T = 100\text{ ms}$)	Detector gate window / pump bin timing
Error Structure	One-sided ($x \notin L \implies Q = 0$)	Zero dark-count ideal photodetector
Feedback Loop	NPN / gate feedback to pump driver	Deutsch CTC self-consistency condition

> **Conclusion:** The optoelectronic decision circuit successfully bridges computational complexity (RP single-trial evaluation), hardware electronic design (NPN thresholding, logic gates, flip-flop latching), optical physics (Fano factor photon statistics), and theoretical physics models (Deutsch CTC feedback fixed points).